Complete at least 2 exercises.

Throughout, $\Bbbk$ is a field and all modules are graded right modules. A graded algebra $A = \bigoplus_{i \geq 0} A_i$ is locally finite-dimensional with $S = A_0$ semisimple; in all explicit examples $S = \bigoplus_i \Bbbk e_i$ with coordinate idempotents e_i.

Exercise 1.

(1) Let $B = T_{\Bbbk}(\Bbbk x \oplus \Bbbk y)/(x^2)$ with $\deg x = \deg y = 1$.

i. Let $a_d = \dim_{\Bbbk} B_d$. Prove that $a_d = a_{d-1} + a_{d-2}$ for $d \geq 2$ and $h_B(t) = \dfrac{1+t}{1-t-t^2}$.

ii. Compute $B^!$ (with $X = x^*$, $Y = y^*$).

iii. Find a homogeneous basis of $B^!$ and compute $h_{B^!}(t)$.

(2) For $q \in \Bbbk$ let $C_q = \Bbbk\langle x, y\rangle/(x^2,\ y^2,\ xy + qyx)$ with $\deg x = \deg y = 1$.

i. Prove that $C_q \cong C_{q'}$ as graded algebras if and only if $q = q'$ or $qq' = 1$.
Hint. A graded isomorphism is determined by a linear bijection of the degree-1 parts and preserves $\{\Bbbk u \mid u^2 = 0\}$. You may use that $\{1, x, y, yx\}$ is a basis of C_q and that $u^2 = \alpha\beta(1-q)\, yx$ for $u = \alpha x + \beta y$.

ii. Deduce that $C_1 \ncong C_{-1}$ if $\operatorname{char} \Bbbk \neq 2$.

iii. Compute a quadratic presentation of $C_q^!$ (with $X = x^*$, $Y = y^*$).

iv. Compute $h_{C_q^!}(t)$.

v. Compare $C_0^!$ with $C_1^!$.

(3) Let $P_\bullet \to M$ be a projective resolution of an ordinary (ungraded) module with differentials d_h and let $K_1 = \operatorname{Ker} d_0$.

i. Prove that

$$\operatorname{Ext}^1_A(M, N) \cong \operatorname{coker}\big(\operatorname{Hom}_A(P_0, N) \to \operatorname{Hom}_A(K_1, N)\big),$$

where the map is restriction.

ii. Prove that $\operatorname{Ext}^h_A(M, N) \cong \operatorname{Ext}^{h-1}_A(K_1, N)$ for $h \geq 2$.

Exercise 2.

(1) Let $S = \bigoplus_{i=1}^{4} \Bbbk e_i$ and let $V = \Bbbk a \oplus \Bbbk b \oplus \Bbbk c \oplus \Bbbk d$ be the S-bimodule with

$$a \in e_1 V e_2, \qquad b \in e_2 V e_4, \qquad c \in e_1 V e_3, \qquad d \in e_3 V e_4,$$

and $e_i V e_j = 0$ for all other (i, j). For $\lambda \in \Bbbk$ put $A_\lambda = T_S(V)/(ab - \lambda cd)$.

i. Find $R^\perp$ and a presentation of $A_\lambda^!$, for every $\lambda \in \Bbbk$.

ii. Check that $(A_\lambda^!)^! \cong A_\lambda$.

iii. Prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are either both zero or both non-zero.

iv. Check that

$$h_{A_\lambda}(t) = \begin{pmatrix} 1 & t & t & t^2 \\ 0 & 1 & 0 & t \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

v. Find the minimal graded projective resolution of each simple module $e_i S$.

vi. Deduce that A_λ is Koszul for every $\lambda \in \Bbbk$.

vii. Compute $h_{A_\lambda^!}(t)$.

viii. Verify that $h_{A_\lambda}(t)\, h_{A_\lambda^!}(-t)^{\mathsf{T}} = I_4$.

(2) Let A be a graded algebra with $A_0 = \Bbbk$ and let M be a bounded below locally finite-dimensional graded A-module whose minimal graded projective resolution $P_\bullet \to M$ has $P_h = 0$ for $h \gg 0$. Prove that

$$h_M(t) = h_A(t) \cdot \sum_{h \geq 0} (-1)^h \sum_{i \in \mathbb{Z}} \dim_{\Bbbk} \operatorname{ext}_A^h(M, \Bbbk(-i))\, t^i.$$

(3) Let M be a graded right module with $M = M_m A$ for some $m \in \mathbb{Z}$, bounded below and locally finite-dimensional, with minimal graded projective resolution $P_\bullet \to M$.

i. Prove that $(P_h)_i = 0$ for $i < m + h$.

ii. Deduce that $\operatorname{ext}^h(M, L(j)) = 0$ for $j > -m - h$ and every simple graded S-module L in degree 0.

Exercise 3.

(1) Let $n \geq 3$ and $Z \subseteq \{1, \dots, n-2\}$, and let $A_Z = \Bbbk A_n/(e_{k,k+2} \mid k \in Z)$ be the quotient of the algebra of upper triangular $n \times n$ matrices, with $\deg e_{ij} = j - i$.

i. Check that $h_{A_Z}(t)_{ij} = t^{j-i}$ if $i \leq j$ and there is no $k \in Z$ with $i \leq k \leq j-2$, and 0 otherwise.

ii. For each i, find the minimal graded projective resolution of $e_{ii}S$.
Hint. Let $\ell_i \geq 0$ be maximal with $i, i+1, \dots, i+\ell_i-1 \in Z$. Show that $P_h = e_{i+h,i+h}A(-h)$.

iii. Deduce that A_Z is Koszul.

iv. Compute $\dim_{\Bbbk} \operatorname{ext}^h(e_{ii}S, e_{jj}S(m))$ for all h, j, m, with ℓ_i as in the hint of (b).

v. Let $n = 4$ and $Z = \{1\}$. Then $A_Z = T_S(V)/(R)$ with $S = \bigoplus_i \Bbbk e_{ii}$, $V = \Bbbk e_{12} \oplus \Bbbk e_{23} \oplus \Bbbk e_{34}$ and $R = \Bbbk\,(e_{12} \otimes e_{23})$. Compute $R^{\perp}$ and a presentation of $A_Z^!$.

vi. Find a basis of $A_Z^!$ and compute $h_{A_Z^!}(t)$.

vii. Verify that $h_{A_Z}(t)\, h_{A_Z^!}(-t)^{\mathsf{T}} = I_4$.

(2) Let A be a Koszul algebra with $S = \bigoplus_{i=1}^{n} \Bbbk e_i$, and let $b_h^{(i)}$ be the dimension vector of $\operatorname{grtop} P_h$ for the minimal graded projective resolution of e_iS; let B_h be the matrix with rows $b_h^{(i)}$.

i. Prove that
$$\sum_{h \geq 0} (-1)^h t^h B_h \cdot h_A(t) = I_n.$$

ii. Deduce that the coefficient of t^h in $h_A(t)^{-1}$ has entries of sign $(-1)^h$ (or zero).

(3) Let $A = \Bbbk\langle x, y\rangle/(yx,\ x^2 + y^2)$ with $\deg x = \deg y = 1$.

i. Show that $x^3 = 0$ and $y^3 = 0$, and deduce that $A_d = 0$ for $d \geq 3$ and $h_A(t) = 1 + 2t + 2t^2$.
Hint. $x^3 = x \cdot x^2 = -xy^2$ and $x^3 = x^2 \cdot x = -y^2x = -y(yx)$, and similarly for the other words of length 3.

ii. Using part (2), prove that A is not Koszul, although it is quadratic.

Exercise 4.

(1) Let $n \geq 2$ and $A = \Bbbk[x]/(x^n)$ with $\deg x = 1$.
   i. Find the minimal graded projective resolution of $\Bbbk = A/A_+$.
   ii. Compute $\operatorname{ext}^h_A(\Bbbk, \Bbbk(j))$ for all h, j.
   iii. Let $1 \leq r < n$ and $M = A/x^r A$. Find the minimal graded projective resolution of M.
   iv. Compute $\operatorname{ext}^h_A(M, \Bbbk(j))$ for all h, j.

(2) Let M be a bounded below locally finite-dimensional graded module with minimal graded projective resolution $P_\bullet \to M$, and let $n \geq 0$. Prove that the following are equivalent:
   (a) $P_h = 0$ for all $h > n$;
   (b) $\operatorname{ext}^{n+1}(M, L(j)) = 0$ for every simple graded S-module L in degree 0 and every $j \in \mathbb{Z}$.

   Hint. Use the description of $\operatorname{ext}^h(M, L(j))$ via $\operatorname{grtop} P_h$ and the graded Nakayama lemma.

(3) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and $V = \Bbbk a \oplus \Bbbk b$ with $a \in e_1 V e_2$, $b \in e_2 V e_1$, and $A = T_S(V)/(a \otimes b)$.
   i. Check that $h_A(t) = \begin{pmatrix} 1 & t \\ t & 1 + t^2 \end{pmatrix}$.
   ii. Find the minimal graded projective resolutions of $e_1 S$ and $e_2 S$.
   iii. Deduce that A is Koszul.
   iv. Compute a presentation of $A^!$.
   v. Find a basis of $A^!$ and compute $h_{A^!}(t)$.
   vi. Verify that $h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I_2$.